

Program: Diploma in Fire Technology and Safety	
Course Code: 3467	Course Title: Mechanical Workshop III
Semester: 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide practice to the students on machine tools like lathe, shaper, etc. so that they can acquire skills for performing operations like plain turning, taper turning, planning etc.
- To provide practice on fitting operations and impart skills on handling various measuring instruments
- To provide practice on different welding operations and impart skills on fabrication works on sheet metals
- To Introduce the casting process through foundry practice and impart the skill for carrying operations on smithy

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Manufacturing processes		Workshop practice	1 & 2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Perform machining operations on lathe and shaper	10	Applying
CO2	Apply technical skill to practice fitting operations and use of various gauges	10	Applying
CO3	Perform fabrication works by making semi-permanent joints in metal sheets and Practice welding operations.	11	Applying

CO4	Apply technical skill to perform smithy and foundry work	10	Applying
	Lab Exam	4	

*Outcomes will be achieved after completing the 2 courses, Mechanical workshop III in third semester (45 hours) and Mechanical workshop IV in Fourth semester (45 hours)

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2		2		3			2
CO3			2	3			
CO4	2			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Perform various operations on lathe/shaping machines		
M1.01	Lathe practice	6	Applying
M1.02	Shaper Practice	4	Applying
CO2	Apply technical skill to practice fitting operations and use of various gauges		
M2.01	Fitting practice	10	Understanding
	Lab Exam I	2	
CO3	Perform fabrication works by making semi- permanent joints in metal sheets and Practice welding operations		
M3.01	Fabrication using sheet metal and various sections of metal and nonmetals & alloys..	5	Applying
M3.02	Welding practice	6	Applying
CO4	Apply technical skill to perform smithy and foundry work		

M4.01	Smithy practice.	5	Applying
M4.02	Foundry practice	5	Applying
	Lab Exam – II	2	

Text / Reference:

T/R	Book Title/Author
T1	Manufacturing process – Myro N Begman, 5 th edition, Tata McGraw Hill, New Delhi
T2	Production Technology – HMT, 18th edition, Tata McGraw Hill, New Delhi
R1	Elements of Workshop Technology (Volume I & II) – HajraChowdry&Bhattacharya, Media Promoters, 11th Edition, 2007
R2	Production technology---By R.K. JAIN, Khanna publishers
R3	Manufacturing technology—By R.K. RAJPUT, Laxmi publications (P) Ltd

Online Resources:

Sl No	Website Link
1	https://nptel.ac.in/course.html
2	https://www.metalcuttingvision.com/machine-shop-tutorial/
3	http://www.thatlazymachinist.com/free-training.html
4	www.nptel.ac.in/courses/mechanical